

Algorithm Design Paradigms

Handout on Asymptotics

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This handout concerns solving for Asymptotics.

1. $T(n) = T(\sqrt{n}) + O(n)$
2. $T(n) = 6T\left(\frac{n}{3}\right) + O(n^2)$
3. $T(n) = 4T\left(\frac{n}{5}\right) + O(n)$
4. $T(n) = 3T(\sqrt{n}) + \log(n)$

In each of the following situations, indicate whether $f = O(g)$, or $g = O(f)$, or both (in which case $f = \Theta(g)$) or neither (is that even possible?!)

5. $f(n) = n \log n$ and $g(n) = 10n \log(10n)$.
6. $f(n) = n^{1.01}$ and $g(n) = n \log^2 n$.
7. $f(n) = (\log n)^{\log n}$ and $g(n) = \frac{n}{\log n}$
8. $f(n) = \log(n!)$ and $g(n) = n \log(n)$
9. For a given $k < n$, $f(n) = \binom{n}{k}$ and $g(n) = n^k$
10. $f(n) = n$ and $g(n) = n^{1+\sin n}$

Assume that $T(n) = 1$ for $n < N$ for some suitable N .

11. $T(n) = \sum_{i=1}^{n-1} T(i) + 1$
12. $T(n) = T\left(\frac{7n}{10}\right) + T\left(\frac{n}{5}\right) + O(n)$
13. $T(n) = T(\log(n)) + n$
14. $T(n) = T(n - \sqrt{n}) + 1$
15. $T(n) = T(n - 1) + n^k$ for $k > 1$, k is a natural number.